# Karthik Bhattaram

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## Education

**University of California, Santa Barbara** — Santa Barbara
*Master of Science in Computer Science* — *March 2026 to June. 2027*

- Courses: Graph Neural Networks, Classic Papers in Security, Frontier LLM and Agent Capabilties
- Research Area: Network Security, mentored by Christopher Kruegel and Giovanni Vigna

**University of California, Santa Barbara** — Santa Barbara
*Bachelor of Science in Computer Science* — *Sept. 2022 to Mar. 2026*

- Courses: Computer Architecture, Machine Learning for Networking, Internet Computing
- Honors: Regents Scholar, College of Engineering Honors, USACO Silver Divison

## Experience

**Software Engineer Intern** — June 2025 – Sept. 2025
*Arista Networks* — *Santa Clara, California*

- Accelerated Broadcom switching ASIC diagnostics by orders of magnitude using custom C++ utilities
- Developed an interprocess communication tool for Microsoft and OpenAI datacenters

**Machine Learning Research Intern** — Sept. 2024 – June 2025
*University of California, Santa Barbara* — *Santa Barbara, California*

- Published a paper titled "Leveraging Large Language Models to Contextualize Network Measurements"
- Aggregated FCC data into a SQL database to improve model context using retrieval-augmented generation

**Software Engineer Intern** — July 2024 – Sep. 2024
*Cadence Design Systems / OpenEye Scientific* — *Santa Fe, New Mexico*

- Wrote AWS-based package to benchmark database compression algorithms that saved terabytes of server space
- Updated CUDA C++ software to incorporate new language features to match competing industry solutions

**Hardware Architecture Intern** — June 2023 – Sept. 2023
*Cadence Design Systems* — *San Jose, California*

- Implemented a Python automation tool to generate memory wrappers for use on a hardware emulation platform
- Collected foundry provided memories written in System Verilog to be used as templates for this tool

**Hardware Verification Intern** — June 2021 – Aug. 2022
*Axiado Corporation* — *San Jose, California*

- Developed a Python package to create network packet flows for use in hardware simulation and emulation
- Created and executed a test plan for a hardware firewall's parser module and discovered multiple bugs

## Projects

**Graph Fine Tuning** | *Python, PyTorch, Graph Algorithms* — Jan 2026 – March 2026

- Adapted Meta's Llama 3.1 model to process graph generation tasks using parameter-efficient fine tuning
- Implemented an experimental pipeline to compare zero-shot, Supervised Fine Tuning, and Direct Preference Optimization performance
- Achieved 85% improvement in graph correctness (based on graph metrics) over the zero shot model

## Technical Skills

**Certificates**: Enterprise Security Professional, Python for Data Science and AI
**Networking Technologies**: Scapy, Wireshark, TCP/IP, Ethernet
**Languages**: Python, C/C++, Java, JavaScript, Shell, Verilog
**Developer Tools**: Git, Visual Studio Code, Vim, Emacs
**Libraries**: CUDA, Pandas, NumPy, PyTest, Black, Matplotlib